Linear Algebra
Homework 1
Deadline: Thursday October 9th 2025 23:59

The augmented matrix of a linear system has been reduced by row operations to the form shown. In each case, continue the appropriate row operations and describe the solution set of the original system.

$x_1 - 5 = -9$

1. $\begin{bmatrix} 1 & 1 & 5 & 0 \\ 0 & 1 & 9 & 0 \\ 0 & 0 & 7 & -7 \end{bmatrix} \times\frac{1}{7} \sim \begin{bmatrix} 1 & 1 & 5 & 0 \\ 0 & 1 & 9 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \times(-9) \sim \begin{bmatrix} 1 & 1 & 5 & 0 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & -1 \end{bmatrix} \times(-1) \sim \begin{bmatrix} 1 & 0 & 5 & -9 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & -1 \end{bmatrix} \Rightarrow \begin{cases} x_1 = -4 \\ x_2 = 9 \\ x_3 = -1 \end{cases}$ #

2. $\begin{bmatrix} 1 & -2 & 0 & 3 & 0 \\ 0 & 1 & 0 & -4 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \times 2 \sim \begin{bmatrix} 1 & 0 & 0 & -5 & 0 \\ 0 & 1 & 0 & -4 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 - 5x_4 = 0 \\ x_2 - 4x_4 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases}$ #

Solve the following systems.

3. $$\begin{aligned} x_1 - 3x_2 + 4x_3 &= -4 \\ 3x_1 - 7x_2 + 7x_3 &= -8 \\ -4x_1 + 6x_2 + 2x_3 &= 4 \end{aligned}$$

4. $$\begin{aligned} x_1 - 3x_2 \qquad &= 5 \\ -x_1 + x_2 + 5x_3 &= 2 \\ x_2 + x_3 &= 0 \end{aligned}$$

3.

$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ 3 & -7 & 7 & -8 \\ -4 & 6 & 2 & 4 \end{bmatrix} \times\frac{1}{2} \sim \begin{bmatrix} 1 & -3 & 4 & -4 \\ 3 & -7 & 7 & -8 \\ -2 & 3 & 1 & 2 \end{bmatrix} \times 2 \quad \times(-3)$$

$$\sim \begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 2 & -5 & 4 \\ 0 & -3 & 9 & -6 \end{bmatrix} \times\frac{-1}{3} \sim \begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 2 & -5 & 4 \\ 0 & 1 & -3 & 2 \end{bmatrix} \times(-2)$$

$$\sim \begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & -3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \times 3$$

$$\sim \begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \times 3 \sim \begin{bmatrix} 1 & 0 & 4 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 = 2 \\ x_2 = 2 \\ x_3 = 0 \end{cases}$$ #

Determine if the system is consistent.

~~5.~~ $$\begin{aligned} x_1 \qquad\qquad - 2x_4 &= -3 \\ 2x_2 + 2x_3 \qquad &= 0 \\ x_3 + 3x_4 &= 1 \\ 2x_1 + 3x_2 + 2x_3 + x_4 &= 5 \end{aligned}$$

4.

$$\begin{bmatrix} 1 & -3 & 0 & 5 \\ -1 & 1 & 5 & 2 \\ 0 & 1 & 1 & 0 \end{bmatrix} \times 1 \sim \begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & -2 & 5 & 7 \\ 0 & 1 & 1 & 0 \end{bmatrix} \times 2 \sim \begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & 0 & 7 & 7 \\ 0 & 1 & 1 & 0 \end{bmatrix} \times \tfrac{1}{7}$$

$$\sim \begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \times 3 \sim \begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \times(-1) \quad \times(-3)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \Rightarrow \begin{cases} x_1 = 2 \\ x_2 = -1 \\ x_3 = 1 \end{cases}$$ #

5.

$$\begin{bmatrix} 1 & 0 & 0 & -2 & 3 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 2 & 3 & 2 & 1 & 5 \end{bmatrix} \times \tfrac{1}{2} \quad \times(-2) \sim \begin{bmatrix} 1 & 0 & 0 & -2 & 3 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 3 & 2 & 5 & -1 \end{bmatrix} \times(-3)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & -2 & 3 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & -1 & 5 & -1 \end{bmatrix} \times(-1) \quad \times 1 \sim \begin{bmatrix} 1 & 0 & 0 & -2 & 3 \\ 0 & 1 & 0 & -3 & -1 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 8 & 0 \end{bmatrix} \times \tfrac{1}{8} \sim \begin{bmatrix} 1 & 0 & 0 & -2 & 3 \\ 0 & 1 & 0 & -3 & -1 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \times(-3)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & -2 & 3 \\ 0 & 1 & 0 & -3 & -1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 - 2x_4 = 3 \\ x_2 - 3x_4 = -1 \\ x_3 = 1 \\ x_4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = 3 \\ x_2 = -1 \\ x_3 = 1 \\ x_4 = 0 \end{cases} \Rightarrow \text{consistent}$$ #

Determine which matrices are in reduced echelon form and which others are only in echelon form.

6.

a. $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ b. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ d. $\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

a·b ⇒ reduced echelon form

c ⇒ echelon form

d ⇒ ×

Find the general solutions of the systems whose augmented matrices are given below.

7. $\begin{bmatrix} 1 & 4 & 0 & 7 \\ 2 & 7 & 0 & 11 \end{bmatrix} \times(-2)$

7. $\sim \begin{bmatrix} 1 & 4 & 0 & 7 \\ 0 & -1 & 0 & -3 \end{bmatrix}$

$\begin{cases} x_1 = -5 \\ x_2 = 3 \\ x_3 \text{ is free variable} \end{cases}$ #

8. $\begin{bmatrix} 1 & -2 & -1 & 3 \\ 3 & -6 & -2 & 2 \end{bmatrix} \times(-3)$

9. $\begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ -1 & 7 & -4 & 2 & 7 \end{bmatrix}$

8. $\sim \begin{bmatrix} 1 & -2 & -1 & 3 \\ 0 & 0 & 1 & -7 \end{bmatrix} \times 1$

$\sim \begin{bmatrix} 1 & -2 & 0 & -4 \\ 0 & 0 & 1 & -7 \end{bmatrix}$

$\rightarrow \begin{cases} x_1 - 2x_2 = -4 \\ x_3 = -7 \end{cases}$

$\rightarrow \begin{cases} x_1 = -4 + 2x_2 \\ x_2 \text{ is free variable} \\ x_3 = -7 \end{cases}$ #

Suppose each matrix represents the augmented matrix for a system of linear equations. In each case determine if the system is consistent. If the system is consistent, determine if the solution is unique.

10.

a. $\begin{bmatrix} \blacksquare & * & * \\ 0 & \blacksquare & * \\ 0 & 0 & 0 \end{bmatrix}$

b. $\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$

9. $\begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ -1 & 7 & -4 & 2 & 7 \end{bmatrix} \times 1$

$\sim \begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & -4 & 8 & 12 \end{bmatrix} \times \frac{1}{4}$

$\sim \begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & -1 & 2 & 3 \end{bmatrix} \times 1$

$\begin{cases} x_1 - 7x_2 + 6x_4 = 5 \\ x_3 - 2x_4 = -3 \end{cases}$

$\sim \begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\Rightarrow \begin{cases} x_1 - 7x_2 = 5 - 6x_4 \\ x_3 = -3 + 2x_4 \\ x_4 \text{ is free} \end{cases}$ #

10.

a.

$$\begin{bmatrix} \blacksquare & * & * \\ 0 & \blacksquare & * \\ 0 & 0 & 0 \end{bmatrix} \times\left(-\frac{*}{\blacksquare}\right) \sim \begin{bmatrix} \blacksquare & 0 & * - \frac{*^2}{\blacksquare} \\ 0 & \blacksquare & * \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 = * - \frac{*^2}{\blacksquare} \\ x_2 = * \end{cases}$$

Ans: consistent, $x_1 = * - \frac{*^2}{\blacksquare}$, $x_2 = *$

Yes

b.

$$\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare & * \end{bmatrix} \times\left(-\frac{*}{\blacksquare}\right)$$

$\blacksquare x_4 = *$

$x_4 = \frac{*}{\blacksquare}$

$$\sim \begin{bmatrix} \blacksquare & * & * & 0 & *^2 - \frac{*^2}{\blacksquare} \\ 0 & 0 & \blacksquare & 0 & *^2 - \frac{*^2}{\blacksquare} \\ 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

Ans: consistent, but not unique